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Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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Scope Statement

This paper presents the Triune-Intelligence Smart Engineering-VALORIZE (TISE-VALORIZE) Framework, which brings together human, artificial, and natural intelligence in the Perceive-Understand-Design-Act-Learn (PUDAL) cycle and a set of value engines (PSKVE) for digital engineering education. An experimental investigation including 138 students demonstrated a significant rise in their academic scores (M=94.98), accompanied with a reduction in the score range by as much as 84% in comparison to the control group. This suggests that it was more fair and useful to learn. Value Co-Creation, collaborative communities of practice, and AI integration are all in line with the digital learning innovation trends that Frontiers in Education sees, especially in the Digital Learning Innovations area. This research may add to the body of work on AI-based teaching and shared values by providing empirical data and a solid theoretical background. This will help the digitalization agenda of higher education.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

Credit Author Statement

Armein Z. R. Langi: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **I Gusti Bagus Baskara Nugraha**: Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **Meliana Christianti Johan**: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Parlindungan Sinaga**: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft. **Radiant Victor Imbar**: Conceptualization, Data curation, Formal Analysis, Methodology, Resources, Supervision, Validation, Visualization, Writing – review & editing.

Keywords

artificial intelligence, Cognitive Load Theory, Engineering Education, Professional identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

Abstract

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AI can automate technical activities, but it can't teach skills that only people have. This study proposes the TISE-VALORIZE framework and presents a preliminary empirical examination of its association with student performance outcomes in digital engineering education. A posttest-only quasi-experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering students: one intervention group implementing TISE-VALORIZE ($n = 53$) and two control groups receiving conventional instruction ($n = 42$; $n = 43$). Student performance was evaluated through Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention group showed higher mean scores ($M = 94.98$, $SD = 7.21$) and lower score variance ($Var = 52.01$) than Control 1 ($M = 88.70$, $SD = 18.33$, $Var = 335.78$) and Control 2 ($M = 86.54$, $SD = 17.05$, $Var = 290.98$). A one-way ANOVA indicated a significant group difference, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$, and Tukey post-hoc analysis showed a significant difference between the intervention group and Control 2, while the difference between the intervention group and Control 1 did not reach significance. These preliminary results suggest that the framework may support more consistent learning outcomes. However, direct measures of cognitive load, motivation, and related psychological processes were not included in the present study. These findings provide preliminary support for the framework's potential to improve performance consistency, while larger-scale studies with direct measures of cognitive and motivational processes are needed.

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In this study, we used Canva as a drawing application to create Figure 1 and also used Grammarly to check the grammar and writing of the manuscript..

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Abstract

AI can automate technical activities, but it can't teach skills that only people have. This study proposes the TISE-VALORIZE framework and presents a preliminary empirical examination of its association with student performance outcomes in digital engineering education. A posttest-only quasi-experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering students: one intervention group implementing TISE-VALORIZE ($n = 53$) and two control groups receiving conventional instruction ($n = 42$; $n = 43$). Student performance was evaluated through Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention group showed higher mean scores ($M = 94.98$, $SD = 7.21$) and lower score variance ($Var = 52.01$) than Control 1 ($M = 88.70$, $SD = 18.33$, $Var = 335.78$) and Control 2 ($M = 86.54$, $SD = 17.05$, $Var = 290.98$). A one-way ANOVA indicated a significant group difference, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$, and Tukey post-hoc analysis showed a significant difference between the intervention group and Control 2, while the difference between the intervention group and Control 1 did not reach significance. These preliminary results suggest that the framework may support more consistent learning outcomes. However, direct measures of cognitive load, motivation, and related psychological processes were not included in the present study. These findings provide preliminary support for the framework's potential to improve performance consistency, while larger-scale studies with direct measures of cognitive and motivational processes are needed.

1 Introduction

Engineering education is shifting from a content-based approach to competency development (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and industry endures, even among high-achieving graduates: analyses of software engineering practices consistently reveal a misalignment between the focus of higher education programs and the requirements of practitioners in professional settings, including both technical and professional aspects (Garousi, Giray and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and frameworks are being created to test preparedness and include digital transformation (Imbar *et al.*, 2025). Engineering education has long emphasized foundational knowledge, routine problem solving, and performance assessment. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

Learning requires learners to organize information meaningfully and integrate it into schemas stored in long-term memory (Sweller, Van Merrienboer and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. Unless actively rehearsed, such information typically decays within 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus and Sweller, (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

The theoretical framework delineated below aims to actualize this possibility. It is based on well-known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and research on expertise) and considers the realities of modern technology (AI integration, digital tools, and participatory design). Its goal is to create professionals who can lead and make a difference in a world that is changing quickly. In this preliminary study, we empirically evaluate only the performance-related dimension of the framework, while the motivational, cognitive-load, transfer, and professional-identity mechanisms remain theoretical propositions requiring direct measurement in future work. The framework draws on Cognitive Load Theory, Self-Determination Theory, and professional identity formation because these perspectives address three complementary dimensions of engineering education. Cognitive Load Theory explains how structured learning design and external representations may support schema construction and reduce unnecessary processing demands. Self-Determination Theory explains how autonomy, competence, and relatedness may support sustained engagement in learning activities. Professional identity formation helps explain why professional simulation and value-oriented participation matter in engineering education beyond content mastery alone. Together, these perspectives provide the theoretical basis for the TISE-VALORIZE framework and for its expected influence on both learning processes and performance outcomes.

In the present preliminary empirical study, two hypotheses were examined directly:

- H1. Students in the TISE-VALORIZE group will achieve higher Structured Academic Activities scores than students in the conventional instruction groups.
- H2. Students in the TISE-VALORIZE group will show lower score dispersion than students in the conventional instruction groups.

In addition, the framework proposes several secondary hypotheses for future empirical testing:

- H3. The framework may reduce perceived cognitive load through structured knowledge mapping and scaffolding.
- H4. The framework may support intrinsic motivation and learning agency through autonomy-supportive and contribution-oriented learning activities.
- H5. The framework may support stronger professional identity formation and learning transfer through professional simulation and value-oriented participation.
- H6. The framework may increase the quality and reuse value of student-produced artefacts within the learning community.

Because the present study did not include direct measures of cognitive load, motivation, professional identity, or transfer, only H1 and H2 are evaluated empirically in this paper.

2 Materials and Methods

2.1 Theoretical Foundations: Merging Cognitive Science and Pedagogy

2.1.1 Fundamental Ideas About Education

The framework is based on several interconnected ideas that are meant to meet the needs of the AI future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical information, and students learn it. VOE alters this goal. Higher education should teach how to be good people, how to be professionals, and how to give back to society. Students learn how to develop code and why technological choices might hurt people or the environment, for example. Second, Value Co-Creation (VCC). This word originates from marketing and service writing and talks about how businesses and customers work together to produce value. In education, VCC means that students do not merely receive knowledge passively; they actively contribute to creating learning experiences that matter. They collaborate with educators and technology to develop resources such as knowledge maps, problem-solving tools, and educational aids for their classmates and future learners. In this way, students become co-creators instead than merely consumers of knowledge (Prahalad and Ramaswamy, 2004; Grönroos and Voima, 2013; Robayo-Pinzon *et al.*, 2023)

Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share, and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023). It works well with the focus of VOE and VCC. Collaborative Learning (CL) is a strategy in which a small group of students work together to achieve a shared objective or complete an assignment (Ouyang, Xu and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly, Personalized Learning (PL) is a method of teaching that changes based on how fast students learn and how lessons are given to help them achieve their best (Labib, Canós and Penadés, 2017; Deev and Finogeev, 2023; Apirating, 2025).

2.1.2 Psychological and Cognitive Foundations

CKMS-SE combines old ways of thinking about psychology and cognition:

- Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and Bär, 2024; Annamalai and Nasor, 2025).

- Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning process (Alsaffar, Alfayly and Ali, 2022; Apirating, 2025).
- People that have similar aims, interests, and skill levels make up Communities of Practice (CoP). They acquire information collaboratively by active participation and contribution as a cohesive unit (Tsang and Tsui, 2017; Taft, 2019).

2.1.3 Self-Determination Theory and Intrinsic Motivation

CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the motivating factors that sustain student engagement and facilitate profound learning (Ninghardjanti *et al.*, 2025). SDT says that human motivation exists on a scale. At one end is external regulation, where people do things because they feel pressure or get benefits from outside sources. At the other end is integrated regulation, where people do things that are in line with their own beliefs and identities. SDT also makes a distinction between the amount of motivation (how strongly someone feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

Students who still use conventional learning approaches tend to depend on grades or test results as rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the aspiration to make a significant impact, employ more profound learning strategies, maintain persistence despite the lack of external incentives, and experience enhanced well-being and continuous engagement.

2.1.4 Cognitive Load Theory and Schema Construction

New information is first processed in working memory. Working memory has severe capacity and duration constraints (Duran, Zavgorodniaia and Sorva, 2022). Learning is the process of transferring information from working memory to long-term memory, where it is organized into mental schemas that integrate multiple elements into a coherent whole. The creation and automation of such schemas are central to expertise development (Robins, 2022).

This research established the VALORIZE strategy, which aims to design learning that alleviates the three forms of cognitive burden on students and enhances information processing in their working memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert the signal and system domains, which were initially expressed in words and arithmetic equations, into knowledge maps and problem-solving maps.

The following design principle is clear: to make learning as effective as possible, instructional designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for difficult material), and increase germane load, which is the amount of working memory used for productive learning activities (Duran, Zavgorodniaia and Sorva, 2022).

2.1.5 Community of Practice: Professional Identity and Expertise Development

The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that come with a job, is an important part of the framework. Professional identity isn't just something you get by learning new things. It develops over time as you do real professional work, get advice from experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021; Shal *et al.*, 2025).

Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

2.2 The TISE Architecture: Operationalizing Multifaceted Intelligence

In this paper, TISE-VALORIZE refers to the overarching theoretical framework, while CKMS-SE refers to its system-level implementation architecture. The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

2.2.1 Defining the Triune-Intelligence Model

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

2.2.2 The PUDAL Thinking Cycle as A Cognitive Method

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.

Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

PUDAL Phase	Human Intelligence Analysis Task	Artificial Intelligence Analysis Task	Nature Intelligence Analysis Task	Required VALORIZE Artefacts
Perceive (P)	Find out who the most important stakeholders are and what they think about the problem in general.	Define data sources and collection strategies.	Find the most important environmental indicators and physical limits that need to be assessed.	Factual Association Map
Understand (U)	Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is.	Analyze data to identify patterns, correlations, and barriers.	Model the laws of physics and the rules of nature that control the system.	Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart
Decision-making and Planning (D)	Set up ethical guardrails and success metrics that are based on people.	Use optimization algorithms or simulations to evaluate solution options.	Look at your possibilities in light of your resources and how well they work.	Trade-Off Analysis Map (Weighted Graph)
Act-Response (A)	Make user interfaces and service delivery models that people can utilize.	Make algorithms for controlling things, pipelines for processing data, or actions for robots.	Design physical components and artefact mechanisms.	Process Application Map (Flowchart)
Learning-Evaluating (L)	Think about how the answer will affect people's health and equality in the long run.	Think about how the answer will affect people's health and equality in the long run.	Evaluate the non-negotiable final impact of the solution on the environment.	Synthesized Design Map (Network Graph)

2.2.3 The PSKVE engine as a value-creation method

The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets

PSKVE Dimension	TISE-VALORIZE Asset	Dimension of Valued Worth	Example of Artefact Value Creation
Product (P)	Productivity Units (PU)	Economic Value (Productivity and Efficiency)	Writing Python programs to make computations easier or improve designs.
Service (S)	Service Vouchers (SV)	Human-Centered Design (Well-being and Empathy)	Do a thorough study of stakeholders or a map of the user experience.
Knowledge (K)	Knowledge Credits (KC)	Cognitive Depth (Intellectual Capacity)	Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other.
Value (V)	Economic Shares (ES)	Sustainable Exchange (Financial/Social Viability)	Come up with a workable business model or cost-benefit analysis for fair value distribution.
Environment (E)	Sustainability Bonds (SB)	Protective/Ethical Values (Risk and Sustainability)	Do a life cycle evaluation or create systems that really reduce their influence on the environment.

2.3 Operationalization through the Knowledge Marketplace and Knowledge Maps

The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs), flowcharts, and weighted graphs. Git and GitHub make it simple to maintain track of all their assets and make sure that contributions are clear, can be used again, and are straightforward to verify.

2.4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation

2.4.1 Core Principles and the Change from Knowing to Being a Professional

VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and repeating facts and procedures, to professional formation, which is developing the intellectual, ethical, and social traits that make a person a good, responsible professional. This change isn't only a change in words. It has big effects on how teachers and students are responsible for their own learning, how they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who acquire knowledge. The curriculum includes things that need to be taught, and the teachers are professionals who offer that knowledge. How effectively pupils can repeat that material is how success is measured. In a professional development model, on the other hand, students are seen as future professionals who are doing real work. The curriculum includes things that happen in the workplace, and teachers act as mentors to help students grow professionally. Success is based on the ability to add value and contribute.

2.4.2 Professional Simulation as Pedagogical Strategy

The lecturer's job varies a lot. The lecturer is no longer merely a "sage on the stage" who shares information. Now, they are “representative and role models of the profession." This isn't just a shift in words. There are real-world effects: the lecturer demonstrates how to handle difficulties like a pro by stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches students how to be professionals by providing them feedback that goes beyond merely being accurate and looks at aspects like how well they match with professional standards, quality, and depth. Professional values are not only an additional feature of the VALORIZE paradigm; they are the most significant part of any learning activity. When students create control systems, they must not only improve their technical performance but also consider their impact on energy consumption, user-friendliness, and the environment. Learning sends a clear message: being good at something and being ethically responsible are two sides of the same coin.

2.5 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation

The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is a social and motivating framework. This marketplace changes the way students, teachers, and knowledge interact with each other. Instead of just following rules and taking in information, they are now creating and sharing it.

2.5.1 How it works and why

The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as knowledge assignments (such as building a signal-system concept map), and students respond by generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources that future organizations may use to learn from and build on. This sets off a virtuous cycle: when students' effort adds value, they receive higher marks, and their classmates have access to richer learning resources.

Teachers need to ask for artifacts first. They normally put them into groups based on how hard they are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving Models that are good for teaching. It takes a lot of time and thinking to become an expert at these. Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what experts know. People who pass the examinations may "buy" them via a tiered digital currency system. The course's Knowledge Management System releases a few artifacts. These artifacts become permanent resources that present and future students may use and edit, which helps the community learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This makes it clear that performing well in higher education is directly related to being able to undertake important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as benefiting the community), competency via explicit feedback, and autonomy because students may pick what objects they build and how deep their labor is.

2.5.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy

The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study. Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2), students learn the basics by making clear and correct idea maps and systems. When students reach the application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering projects, such as making a filter for an audio system. They look at numerous ways to solve a problem and talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels 4–5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard and messy technical problems. This tiered framework helps students who want to get the most out of their learning naturally go from memorizing to analyzing and synthesizing.

2.6 Roles And Agency: From Consumers to Co-Designers

2.6.1 The Student as Metacognitive Agent and a Source of Value

Students in VALORIZE have a significantly different role than they do in conventional classes. They don't simply passively take in information; they actively learn and add to the group's knowledge. Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive and Problem-Solving Maps, that indicate they understand the material and are useful to the course community. This is not simply a section of the test; it's the most important aspect. Assessment occurs not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own thinking: Students must maintain reflective learning journals that illustrate how they are learning. Using a framework like DAR (Description What Happened; Analysis), students learn to look at their own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness, which is the ability to think about how your brain works. This is important for being able to adjust. Professional Practice: The framework purposefully incorporates industry-standard protocols into student projects. For example, students may utilize version control systems like Git or GitHub to keep track of their learning artifacts, treating the process of learning as a cycle of development. This serves two purposes: it makes a detailed record of the learning process and teaches students to the professional processes they will use in real life. Portfolio Development: Instead of only being graded on their test scores, students build a portfolio of their greatest work, which they typically share on a personal blog or professional website. This portfolio becomes a real asset that shows prospective employers what you can do and immediately helps you advance in your career.

2.6.2 The Educator as Architect, Mentor, and Community Builder

As Learning Architects, teachers plan the whole learning experience, from planning weekly activities and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time and creating social and pedagogical conditions that help students develop their professional identity. This is complicated design job that needs a lot of thought about how people learn. AI takes care of regular tests (such as giving initial feedback and automated scoring), so teachers have more time and energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to show them how specialists solve difficult challenges. They ask deep questions to get students to think more deeply, give them feedback that helps them progress, and help each student on their way to become a skilled and moral professional. This design uses two specialized engines that work together to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine" that pushes constant improvement and change. It is also a PSKVE engine that controls and improves Shared Value Creation (SVC) by making sure that the value provided extends beyond individual learning to the creation of community assets that can be verified. The Core Engine, the AI-based PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation (VCC) are the three key parts of this system.

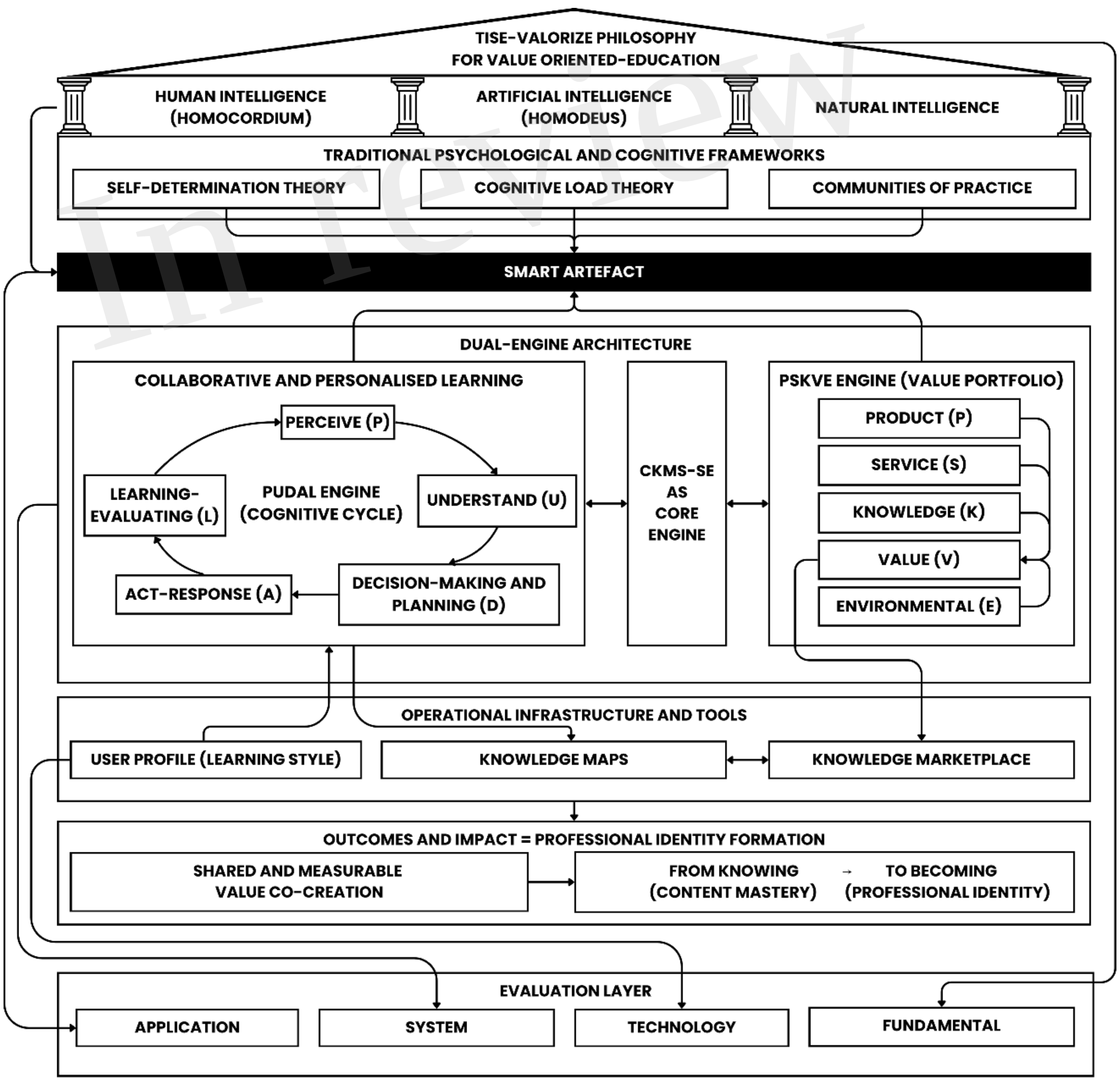

Figure 1. CKMS-SE Architecture

To clarify the empirical scope of the present study, Figure 2 distinguishes between the instructional inputs proposed by the framework, the theoretical mechanisms it draws upon, the performance outcome directly measured in this study, and the broader outcomes proposed for future empirical validation.

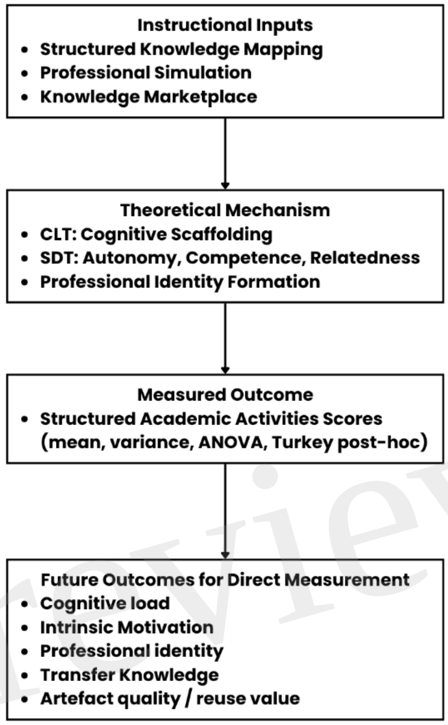

Figure 2. Conceptual scope of the TISE-VALORIZE framework in the present study

3 Preliminary Empirical Validation

3.1 Study Design

The empirical component of this paper employed a posttest-only quasi-experimental design with non-equivalent cross-cohort groups. Random assignment was not feasible because the intervention was implemented in naturally occurring course cohorts across different semesters. Accordingly, the findings should be interpreted as preliminary comparative evidence rather than causal proof.

3.2 Participants and Context

The study involved three cohorts of undergraduate engineering students enrolled in the same course context across different semesters. Control Group 1 (C1) consisted of 42 students from the even semester of 2023/2024 and received conventional instruction. Control Group 2 (C2) consisted of 43 students from the odd semester of 2024/2025 and received conventional instruction. The intervention group (E) consisted of 53 students from the odd semester of 2025/2026 and received instruction using the TISE-VALORIZE framework.

Demographic variables such as age, prior achievement, and access to technology were not systematically collected in a form suitable for retrospective controlled analysis. This limits the extent to which cohort differences can be ruled out statistically and is acknowledged as a limitation of the present study.

3.3 Assessment and Rubric

Student performance was evaluated through Structured Academic Activities using the same Bloom's Taxonomy-aligned rubric across all cohorts. The rubric emphasized conceptual clarity, mapping linkages, procedural reasoning, and integrated synthesis consistent with the expected quality of engineering problem solving. The assessed artefacts included Primitive Knowledge Maps, Problem-Solving Maps, and integrated synthesis artefacts.

Formal psychometric validation and inter-rater reliability statistics for the rubric were not calculated retrospectively in the present study. Therefore, the rubric should be understood as a common instructional assessment instrument rather than a fully validated research-scale measure. This limitation is acknowledged explicitly in the manuscript.

3.4 Statistical Analysis

The analysis included descriptive statistics, one-way ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc comparisons, Levene's test for homogeneity of variances, and Cohen's *d* for pairwise effect size interpretation. Because no pretest or baseline covariates were collected, ANCOVA could not be applied. Likewise, direct measures of cognitive load, intrinsic motivation, professional identity, and transfer were not available in the present dataset.

Before conducting the omnibus comparison, score distributions in each group were examined using the Shapiro–Wilk test. Because the score distributions deviated from normality, a Kruskal–Wallis test was also conducted as a nonparametric robustness check. One-way ANOVA was retained as the primary omnibus test for comparison with prior reporting, followed by Tukey's HSD for pairwise comparisons and Levene's test for variance differences.

3.5 Results

The intervention group showed higher mean performance (M = 94.98, SD = 7.21) than Control 1 (M = 88.70, SD = 18.33) and Control 2 (M = 86.54, SD = 17.05). The intervention group also showed substantially lower variance (Var = 52.01) than both control groups (C1: Var = 335.78; C2: Var = 290.98). These descriptive statistics indicate higher mean scores and lower score dispersion in the intervention group relative to both control groups. This decrease in score dispersion suggests more consistent performance in the intervention group, with fewer students located at the extremes of the score distribution. Table 3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

Group	N	M	SD	Var	Median	IQR	CV (%)	Min	Max
Control 1 (C1)	42	88.70	18.33	335.78	96.66	86.33-99.25	20.7	0.00	100.00
Control 2 (C2)	43	86.54	17.05	290.98	92.91	82.21-96.25	19.7	1.25	98.25
Experiment (E)	53	94.98	7.21	52.01	97.91	93.16-100.00	7.6	65.83	100.00

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation (SD/M × 100); Min/Max = minimum and maximum observed scores.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$. Although statistically significant, the effect size was modest, indicating that group membership was associated with a limited portion of the variance in outcome scores under the conditions of this study. Shapiro–Wilk tests indicated that score distributions in all three groups deviated from normality (Control 1: $W = 0.636$, $p < .001$; Control 2: $W = 0.641$, $p < .001$; Experiment: $W = 0.736$, $p < .001$). Therefore, a Kruskal–Wallis test was conducted as a nonparametric robustness check. The result also indicated a significant group difference, $H(2) = 18.00$, $p < .001$, supporting the conclusion that score distributions differed across groups.

An important result, beyond the higher mean scores, is the substantial reduction in score variability. The intervention group showed a variance of 52.01, compared with 335.78 in Control 1 and 290.98 in Control 2. This pattern indicates lower score dispersion and more consistent performance in the intervention group, whereas the control groups showed substantially greater heterogeneity in outcomes.

Table 4. Summary of ANOVA for Structured Academic Activities scores

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Control (C1)	42	3725.4	88.7	335.7849463
Control (C2)	43	3721.27	86.54116279	290.9826153
Experiment (E)	53	5033.87	94.97867925	52.0050963

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1872.578316	2	936.2891579	4.405265401	0.01401709	3.063203853
Within Groups	28692.71765	135	212.5386493			
Total	30565.29597	137				

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F_crit = critical F-value.

Post-hoc pairwise comparisons using Tukey's Honestly Significant Difference (HSD) test were then conducted to examine which group differences were statistically significant, as shown in Table 5.

Table 5. Comparisons by Pair and Effect Sizes

Comparison	Mean Diff	95% CI	p-value	d (Cohen's)	Interpretation
E versus C1	6.28	[-0.86, 13.42]	.097	0.47	Not significant
E versus C2	8.44	[1.35, 15.53]	.015	0.67	Significant (medium)
C1 versus C2	-2.16	[-9.65, 5.34]	.774	0.12	Not significant

Note: Mean Diff = difference in group means; 95% CI = 95% confidence interval; p = probability value (two-tailed); Cohen's d = standardized effect size. Significant at the α = .05 level.

Figure 3 presents the score distributions across the three groups using a violin plot with box-plot overlay. The intervention group shows a more concentrated distribution in the upper score range and a narrower interquartile spread, whereas the control groups display broader and more dispersed score distributions. This pattern is consistent with the descriptive statistics and variance analysis, which suggest more stable performance in the intervention group under the conditions of this study.

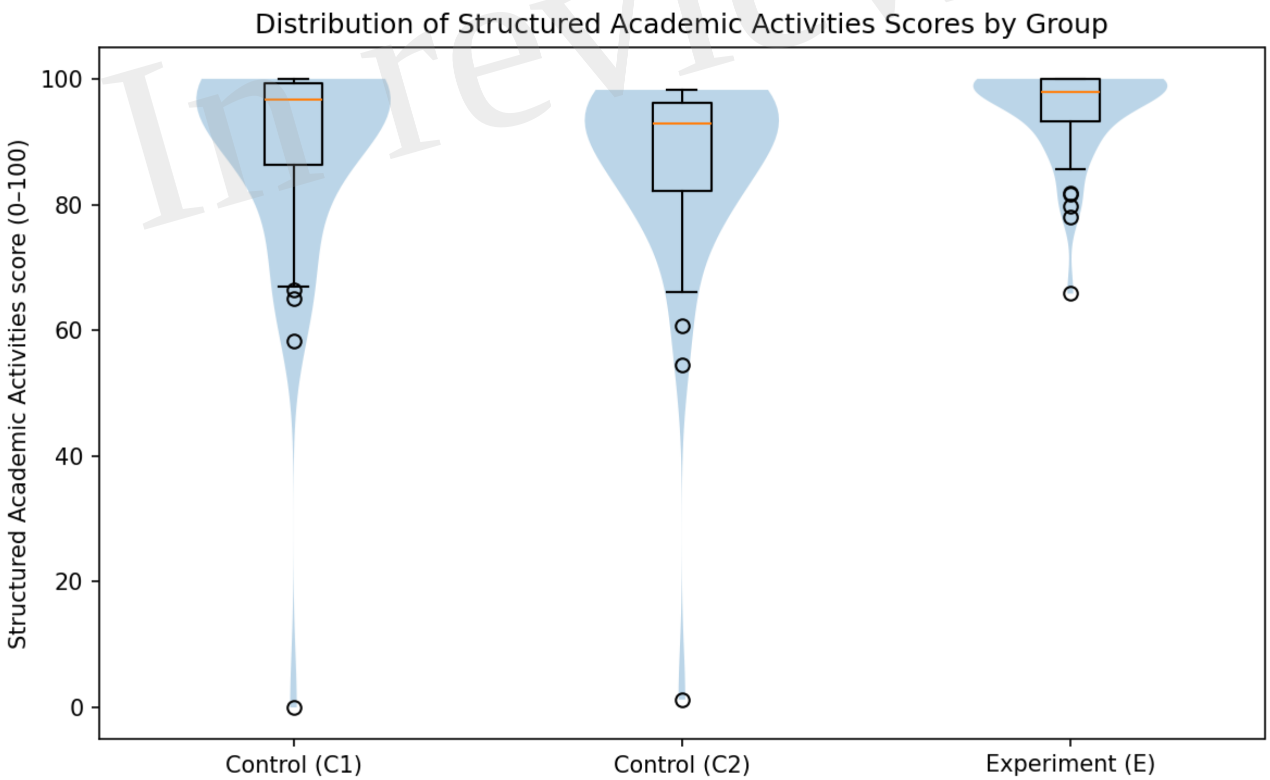

Figure 3. Violin plot with box-plot overlay of Structured Academic Activities scores by group

Post-hoc analysis showed a significant pairwise difference between the Experiment and Control 2 groups (Mean Diff = 8.44, 95% CI = [1.35, 15.53], p = .015, Cohen's d = 0.67), indicating a medium effect size. By contrast, the difference between the Experiment and Control 1 groups did not reach statistical significance (Mean Diff = 6.28, 95% CI = [-0.86, 13.42], p = .097, Cohen's d = 0.47). There was also no significant difference between Control 1 and Control 2 (Mean Diff = -2.16, 95% CI = [-

9.65, 5.34], $p = .774$, Cohen's $d = 0.12$). Levene's test for homogeneity of variances indicated that score variances differed significantly across groups, $F(2, 135) = 5.24$, $p = .006$, supporting the observation that the intervention group showed lower variability than the control groups.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), this pattern may be broadly consistent with the intended scaffolding function of the framework. However, because cognitive load was not directly measured in the present study, no direct claim can be made regarding the framework's effect on cognitive load.

4 Discussion: Theoretical Implications and Addressing Counterarguments

The present study does not directly test all proposed mechanisms of the TISE-VALORIZE framework. Instead, it offers a preliminary empirical indication that the framework may be associated with higher and more consistent performance outcomes, while the proposed motivational, cognitive, and identity-related mechanisms remain to be tested in future studies. Practically, the framework suggests that engineering educators can redesign courses around structured knowledge mapping, professional simulation, and AI-supported feedback to improve consistency of student outcomes.

4.1 Interpretation of Findings

The present study provides preliminary empirical evidence that the TISE-VALORIZE framework may be associated with higher and more consistent performance outcomes in a digital engineering education context. The intervention group showed both higher mean scores and lower score dispersion than the two conventional instruction groups. However, the observed effect size was modest, which suggests that the framework may have contributed to performance differences under the conditions of this study, but it does not account for most of the observed variance.

These findings are broadly consistent with the instructional logic of the framework. In particular, the use of structured knowledge mapping, professional simulation, and contribution-oriented learning activities may have supported more stable performance across students. At the same time, the present study does not directly test the proposed motivational, cognitive-load, or identity-related mechanisms. Therefore, these findings should be interpreted as preliminary performance-based evidence rather than direct confirmation of the full theoretical model.

4.2 Comparison with Prior Work

The present findings are broadly aligned with prior research suggesting that structured scaffolding and meaningful participation can support learning in higher education. From a Cognitive Load Theory perspective, the framework is consistent with the use of structured representations to support schema development. From a Self-Determination Theory perspective, the framework is also consistent with the idea that meaningful contribution, perceived choice, and visible progress may support engagement. However, because these mechanisms were not directly measured here, the present study should be understood as conceptually aligned with prior theory rather than as a direct empirical test of those theories.

4.3 Limitations

This study has several important limitations. First, the design was a posttest-only quasi-experimental comparison with non-equivalent cross-cohort groups rather than a randomized controlled evaluation. As a result, cohort effects and semester-level confounding cannot be ruled out. Second, no pretest or baseline covariates were available, which limits causal interpretation and prevented the use of ANCOVA. Third, the empirical analysis relied on performance outcomes from Structured Academic Activities and did not directly measure cognitive load, intrinsic motivation, professional identity, or transfer. Fourth, demographic variables suitable for controlled comparison were not systematically available. Fifth, Structured Academic Activities were assessed by a single course assessor using the same Bloom's Taxonomy-aligned rubric across cohorts. Formal inter-rater reliability was therefore not applicable. However, no retrospective intra-rater reliability check was conducted, and formal psychometric validation evidence for the rubric was not calculated retrospectively. This limits the extent to which score comparisons can be attributed solely to the intervention. These limitations mean that the findings should be interpreted as preliminary and performance focused.

4.4 Future Directions

Future studies should test the framework using stronger designs, including matched or randomized groups, pretest-posttest comparisons, and larger multi-institutional samples. In addition, future work should include validated instruments for cognitive load, intrinsic motivation, professional identity, and transfer, so that the proposed mechanisms of the framework can be examined directly. Secondary indicators such as artefact quality, marketplace participation, and reuse value may also provide a richer picture of how value-oriented learning operates in practice.

5 Conclusion

This study proposes the TISE-VALORIZE framework as a value-oriented approach to digital engineering education that integrates structured knowledge mapping, professional simulation, and contribution-based learning. In this preliminary posttest-only quasi-experimental comparison, the intervention group showed higher mean performance and lower score dispersion than two conventional instruction groups, although the observed effect size was modest. Practically, the framework suggests a possible direction for redesigning engineering courses around structured representation, authentic participation, and AI-supported feedback. Future research should test the framework with stronger designs and direct measures of cognitive load, motivation, professional identity, and transfer.

6 Additional Requirements

The datasets generated for this study are available on request to the corresponding author.

7 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

8 Author Contributions

All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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Figure 1.JPEG

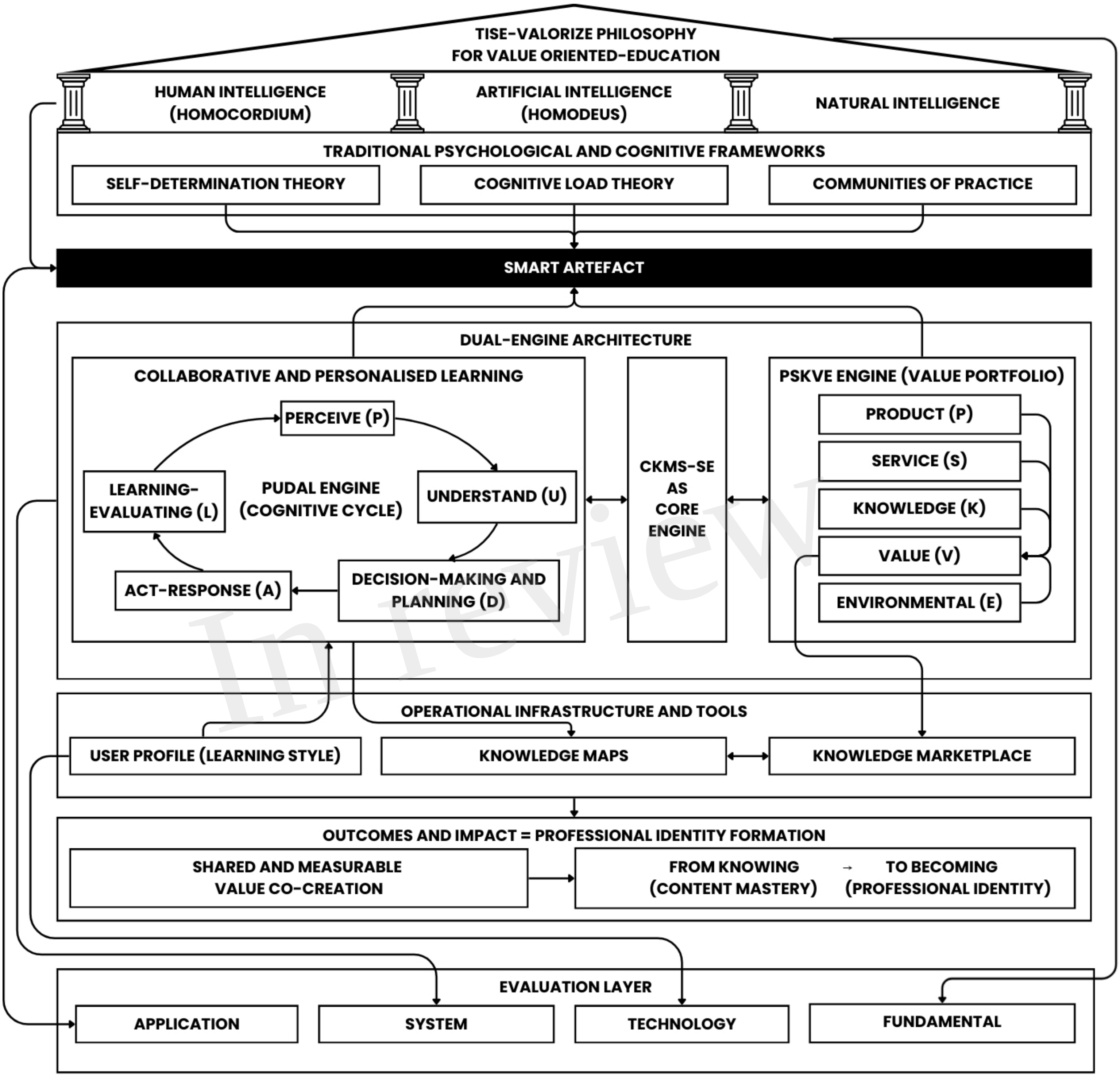

Figure 2.JPEG

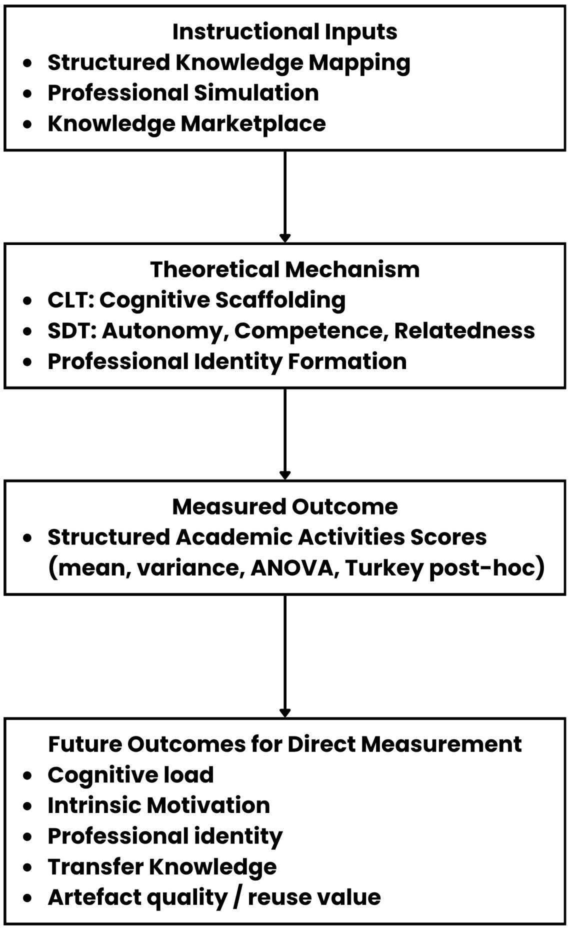

Figure 3.JPEG